Fundamental Limits of Vector Recording

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The hard disk drive (HDD) industry require substantial data storage capacity and Heat-Assisted Magnetic Recording (HAMR) combined with vector recording can be a promising solution. This work investigates the performance limits of vector recording by incorporating reader thermal fluctuations, inter-track interference, synchronization offset, and crosstalk within a unified micromagnetic framework. Reader magnetic noise, modeled using the Landau–Lifshitz–Gilbert equation, results in an SNR reduction of approximately 1–1.5 dB. Comparative analysis with conventional two-dimensional magnetic recording (TDMR) demonstrates that vector recording achieves nearly 3 dB higher peak SNR without aggressive reader scaling. Furthermore, analysis of down-track transition offsets shows that synchronization between adjacent tracks strongly influences signal superposition, SNR, and crosstalk, particularly near the inter-track boundary. The results highlight the robustness and scalability of vector recording for next-generation high-density HAMR systems.

Index Terms—**Areal Density Capacity, HAMR, Reader Design, Read-Head Dimension, Vector Recording.**

I. Introduction

THE DEMAND for data storage is growing exponentially over time due to modern technologies such as artificial intelligence and advanced communications. On the other hand, the hard disk drive (HDD) industry is experiencing a data storage crisis as the existing data storage architectures approach their limits in enhancing areal density capacity (ADC) [1]. In recent times, the HDD industry has shifted from perpendicular magnetic recording (PMR) to heat-assisted magnetic recording (HAMR) to meet this immense demand for data storage. In HAMR systems, ADC improvement is possible by writing smaller bits and narrower tracks. However, continued scaling toward shorter bit lengths and narrower track widths requires aggressive reader miniaturization, which fundamentally limits further gains in bits per inch (BPI) and tracks per inch (TPI). Although two-dimensional magnetic recording (TDMR), as shown in Fig. 1 with ellipse B, improves areal density and relaxes strict reader scaling requirements, it remains insufficient to meet future storage demands [1], [2]. As magnetic recording moves toward ultra-high areal densities, reader noise and inter-track interference increasingly constrain the achievable SNR in TDMR, motivating the need for recording architectures beyond conventional TDMR. Previously, we described vector recording that employs two readers, one conventional reader and one vector reader, to detect magnetic fields parallel and perpendicular to the recording layer, thereby improving performance while relaxing aggressive reader dimension scaling, as shown in Fig. 1 with ellipse A [1]. Furthermore, to overcome vector reader biasing limitations, we introduced alternative reader designs with free-layer bias angles of 45° and 135°, which provide comparable performance through the $X_1 + X_2$ and $X_1 - X_2$ operations, where X_1 and X_2 are readback signal from the 45° and 135° readers [1].

In our earlier study [1], three key factors affecting vector recording performance were not investigated and remain unresolved. First, the simulations included only media noise and neglected reader magnetization noise, which is an important contributor to overall magnetic recording noise. Second, no direct performance comparison was carried out between vector recording and TDMR, where it has been argued that two

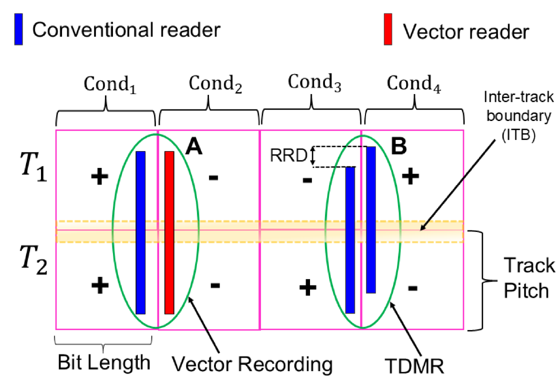

Fig. 1. (a) Readback strategy for vector recording and TDMR.

reduced-size conventional readers can approximate vector recording performance through the operations $S_1 - \beta \times S_2$ or $S_2 - \beta \times S_1$ where S_1 and S_2 are the readback signals from two conventional readers and β is the inter-track coupling coefficient [2]. Finally, the earlier study considered perfectly synchronized writing between adjacent tracks, while synchro nization loss between Track 1 and Track 2 can strongly affect inter-track interference and overall recording performance. In this work, these open questions are investigated through micromagnetic simulations.

II. Effects Of Reader Magnetic Noise

We explicitly model and incorporate reader magnetic noise to evaluate its impact on overall readback performance. First, the power spectral density (PSD) of the reader's free layer is obtained from intrinsic thermal magnetization fluctuations using the Landau–Lifshitz–Gilbert (LLG) equation [1]. The PSD was analyzed for different free-layer bias fields and external field frequencies, revealing that the magnetic fluctu ation behavior is primarily determined by the bias field rather than the excitation frequency. This observation enables the direct incorporation of free-layer thermal noise into the HAMR readback signal modeling framework with a constant bias field. Two recording patterns are considered: same-polarity single-tone bits (sST) and reverse-polarity single-tone bits (RP-sST) between adjacent tracks, resulting in + + (cond_1), − − (cond_2)

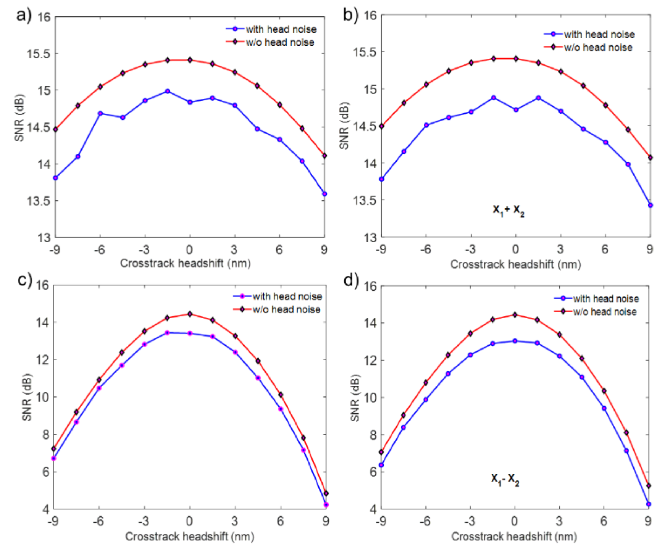

Fig. 2. SNR characteristics as a function of cross-track headshift for different reader designs: (a) R_c with sST, (b) 45° and 135° readers with sST, (c) R_v with RP-sST, and (d) 45° and 135° readers with RP-sST.

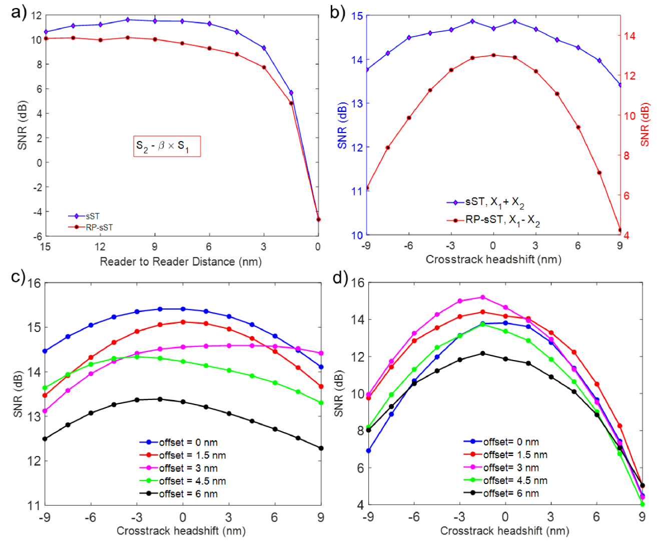

Fig. 3. SNR characteristics for (a) TDMR with 21 nm reader width, (b) vector recording with 25 nm reader width, (c) Rc for sST at different down-track offsets of Track 2 relative to Track 1, and (d) Rv for RP-sST at different down-track offsets of Track 2 relative to Track 1.

and − + ($cond_3$), + − ($cond_4$) magnetizations states, respectively as shown in Fig. 1. Figure 2(a) shows the SNR of the conventional reader (R_c) for the sST pattern, where a strong perpendicular magnetic field produces the maximum positive SNR near the inter-track boundary (ITB) region (-1.5 nm to +1.5 nm) [1]. Including reader thermal fluctuations causes an additional SNR reduction of about 1 dB. Similar performance is achieved using the 45° and 135° reader configurations with the $X_1 + X_2$ operation, as shown in Fig. 2(b). For the RP-sST pattern, the vector reader (Rv) senses the parallel magnetic field and produces a similar SNR response near the ITB region, as shown in Fig. 2(c). Comparable performance is also obtained using the 45° and 135° readers with the $X_1 - X_2$ operation, although an additional SNR loss of about 1.5 dB is observed, as shown in Fig. 2(d). This larger degradation occurs because the opposite-polarity signal components partially cancel, while the uncorrelated reader magnetic noise does not cancel out.

III. TDMR Vs Vector Recording

Fig. 3(a) shows the SNR performance of TDMR with a two-reader configuration (reader width = 21 nm) as a function of reader-to-reader distance (RRD) for both sST and RP-sST patterns. The coupling coefficient β is estimated for each RRD [3] and applied in the $S_2-\beta\times S_1$ operation during readback. At RRD = 0 nm, both readers are positioned at the center of the ITB, resulting in maximum adjacent-track interference. As the RRD increases, the interference decreases and the SNR improves, reaching its maximum at RRD = 9 nm with β = 0.21, where the readers are aligned with the centers of their respective tracks (track pitch = 18 nm). Beyond the optimum RRD, the SNR decreases due to reduced sensitivity to the target-track signal as the readers move farther apart. Using the $S_2 - \beta\times S_1$ operation, TDMR achieves a maximum SNR of 11.5 dB for the sST pattern and 10.1 dB for the RP-sST pattern. In comparison, vector recording using the $X_1 + X_2$ and $X_1 - X_2$ operations achieves peak SNR values of 14.8 dB and 13.2 dB for the sST and RP-sST patterns, respectively, as shown in Fig. 3(b). Therefore, vector recording provides nearly 3 dB higher SNR than TDMR without relying on aggressive reader scaling and inter-track interference cancellation.

IV. Effects Of Write Synchronization

Figures 3(c) and 3(d) show the SNR variation of Rc and Rv with cross-track head shift under different down-track transition offsets between Track 1 and Track 2 for the sST and RP-sST patterns, respectively. From Fig. 3(c), the SNR decreases as the transition offset increases because the temporal overlap of the + + and − − magnetic field components is reduced, weakening constructive signal superposition. A transition offset of around 4–6 nm produces nearly 3 dB SNR degradation for a 20 nm bit length. In contrast, from Fig. 3(d), the SNR initially increases, reaches a maximum near a 3 nm offset, and then decreases with further offset. This behavior originates from curved bit transition, and careful analysis of the writing patterns reveals that an inherent ∼ 3 nm transition shift exists even under synchronized writing conditions. Therefore, introducing a 3 nm down-track offset improves transition alignment and maximizes the SNR, highlighting the strong sensitivity of vector recording to write synchronization. Together, these results highlight the robustness and scalability of vector recording for future high-density HAMR systems.

References

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